

Section: _____ Name: _____

Read the following directions carefully. DO NOT turn to the next page until the exam has started.

Write your name and section number at the top right of this page:

Class	Section Number
Sahifa 8:50	001
Miranda 9:55	002
Sahifa 2:25	003
Sahifa 3:30	004
Miranda 8:50	005

As you complete the exam, write your initials at the top right of each other page.

When the exam start time is called, you may turn the page and begin your exam.
If you need more room, there is a blank page at the end of the exam, or we can give you some scratch paper.

Some multiple choice questions are “Select ONE” while others are “Select ALL that apply”. Pay attention to the question type and only mark one option if it says “Select ONE”. Fill in the circles completely.

If you finish early, you can hand your exam to your instructor or TA and leave early.

Otherwise, stop writing and hand your exam to your instructor or TA when the exam stop time is called.

1. Your zoologist friend is studying three species of zebra. She has gathered data on the physical characteristics of 160 of these zebras and asks for your help analyzing it. Below are 5 example rows from full dataset.

	body_length	shoulder_height	sex	species
	<i><dbl></i>	<i><dbl></i>	<i><chr></i>	<i><chr></i>
1	237	142	F	mountain
2	245	145	M	plains
3	216	128	F	mountain
4	235	139	F	plains
5	282	152	M	grevy's

She sends you the following message about how she wants to graph the data: “I want to make a scatter plot of each zebra’s body length and shoulder height, with a single smooth line showing the trend. I want the color of the points to be different for the different species, and I want all of the points to be slightly transparent in case they overlap”.

- (a) (6 points) Which `ggplot2` graph types should you add to the plot to fulfill her request? **Select ALL that apply.**

- | | |
|--|---|
| ☐ <code>geom_density()</code> | ☒ <code>geom_smooth()</code> |
| ☒ <code>geom_point()</code> | ☐ <code>geom_histogram()</code> |
| ☐ <code>geom_line()</code> | ☐ <code>geom_boxplot()</code> |

- (b) (4 points) For each of the following aesthetics in your plot, identify whether they would need to be global variable, local variable, or constant. **Select ONE each.**

- | | | | |
|-----------|--|---|---|
| • x: | ☒ Global variable | ☐ Local variable | ☐ Constant |
| • y: | ☒ Global variable | ☐ Local variable | ☐ Constant |
| • color: | ☐ Global variable | ☒ Local variable | ☐ Constant |
| • alpha : | ☐ Global variable | ☐ Local variable | ☒ Constant |

2. (3 points) Continue working with the zebra data from problem 1.

	body_length	shoulder_height	sex	species
	<dbl>		<dbl> <chr>	<chr>
1	237	142	F	mountain
2	245	145	M	plains
3	216	128	F	mountain
4	235	139	F	plains
5	282	152	M	grevy's

You want to re-structure the data so that all of the measurements are in a single column called *measurement*, with a column called *type* labeling what is being measured.

	sex	species	type	measurement
	<chr>	<chr>	<chr>	<dbl>
1	M	grevy's	body_length	273
2	M	grevy's	shoulder_height	146
3	M	grevy's	body_length	294
4	M	grevy's	shoulder_height	157
5	M	grevy's	body_length	279
6	M	grevy's	shoulder_height	157

Which line of R code should be used? **Select ONE.**

- ☒ `pivot_longer(body_length:shoulder_height, names_to = "type", values_to = "measurement")`
- ☐ `pivot_wider(body_length:shoulder_height, names_to = "type", values_to = "measurement")`
- ☐ `pivot_longer(names_from = "type", values_from = "measurement")`
- ☐ `pivot_wider(names_from = "type", values_from = "measurement")`

3. (4 points) Continue working with the three species of zebra data from problem 1. Consider the following code which creates a new dataframe, `zebra_summary`.

```
zebra_summary <- zebras %>%
  group_by(species) %>%
  summarize(avgLength = mean(body_length),
            avgHeight = mean(shoulder_height))
```

Which of the following statements are true about `zebra_summary`? **Select ALL that apply.**

- ☒ `zebra_summary` has three rows.
- ☐ `zebra_summary` has two columns.
- ☒ `zebra_summary` has a column called "species".
- ☐ `zebra_summary` has a column called "sex".

4. Consider a standard normal variable $Z \sim N(0, 1)$. The table below lists several lower-tail probabilities of Z .

Z-score	$P(Z \leq -1.5)$	$P(Z \leq -1)$	$P(Z \leq -0.5)$	$P(Z \leq 0.5)$	$P(Z \leq 1)$	$P(Z \leq 1.5)$
<code>pnorm</code>	<code>pnorm(-1.5)</code>	<code>pnorm(-1)</code>	<code>pnorm(-0.5)</code>	<code>pnorm(0.5)</code>	<code>pnorm(1)</code>	<code>pnorm(1.5)</code>
Probability	0.067	0.159	0.309	0.691	0.841	0.933

Use the table of probabilities to answer the questions below.

- (a) (3 points) What is $P(-1 \leq Z \leq 0.5)$? **Select ONE.**

- ☐ 0.691
 ☐ $0.691 + 0.159$
☒ $0.691 - 0.159$
☐ $1 - (0.691 + 0.159)$

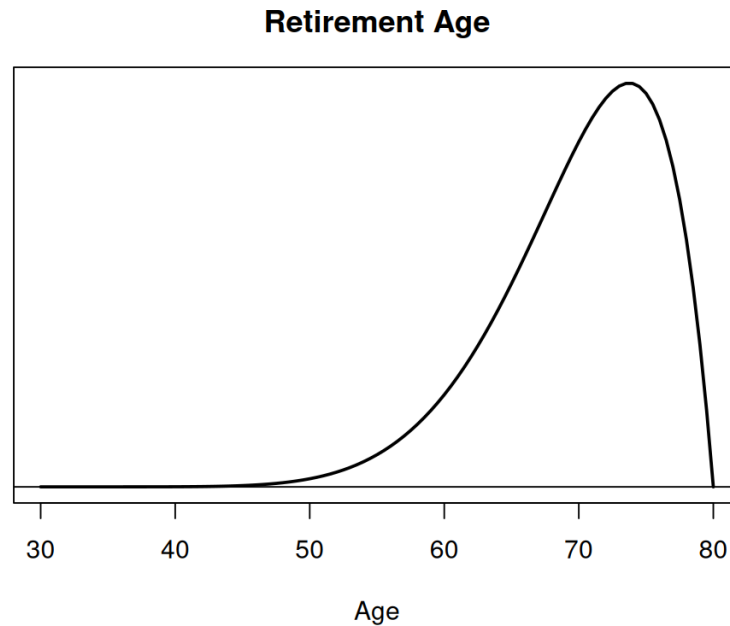
- (b) (3 points) What is $P(Z \leq -1 \text{ OR } Z > 1)$? **Select ONE.**

- ☐ 0
 ☐ 0.691
☐ 0.159
 ☐ 2×0.691
☒ 2×0.159
☐ 1

- (c) (3 points) What value is closest to the 95th percentile of Z ? **Select ONE.**

- ☐ -1.5
 ☐ 0.5
☐ -1
 ☐ 1
☐ -0.5
 ☒ 1.5

5. The continuous random variable X , shown below, represents the population of retirement ages in the US. X is defined on $[30, 80]$ and has expected value 64 and variance 92.16.



- (a) (3 points) **Select ONE.** X will realize to a value greater than 64:
- ☐ less than half the time
 - ☐ close to half the time
 - ☒ more than half the time
- (b) (3 points) **Select ONE.** The sample mean $\bar{X}_{75}$ is calculated by taking the average of 75 random draws from X . $\bar{X}_{75}$ will realize to a value greater than 64:
- ☐ less than half the time
 - ☒ close to half the time
 - ☐ more than half the time
- (c) (5 points) Fill in the blanks to write R code to approximate $P(\bar{X}_{75} \geq 62)$. Your answer should be a function with four arguments.

pnorm (62 , 64 , sqrt(92.16/75) , lower.tail = F)

6. The dataset `skincancer` records the skin cancer mortality (`Mort`) in deaths per 10 million and the latitude (`Lat`) in degrees of the 48 contiguous US states. For example, Florida has latitude 28.0, and Wisconsin has latitude 44.5.

Below is a partial `summary` output for the linear model for skin cancer mortality (`Y`) versus latitude (`X`)

```
skincancer_mod <- lm(Mort ~ Lat, data = skincancer)
summary(skincancer_mod)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	388.312	23.767	16.338	< 2e-16 ***
Lat	-5.966	0.597	-9.994	4.15e-13 ***

- (a) (4 points) Let β_1 be the true linear relationship between skin cancer mortality and latitude. Identify the test statistic and degrees of freedom for the following hypotheses:

$$H_0 : \beta_1 = -4 \quad \text{versus} \quad H_A : \beta_1 \neq -4$$

The test statistic is given by

$$t_{obs} = \frac{-5.966 - 4}{0.597}$$

with $n - 2 = 46$ degrees of freedom.

- (b) (3 points) **Select ONE.** We know that that the p-value for the test in (a) must be:
- ☐ Less than 4.15×10^{-13}
 - ☐ Equal to 4.15×10^{-13}
 - ☒ Greater than 4.15×10^{-13}
- (c) (4 points) A 95% confidence interval for the average mortality at a latitude of 35 is reported as (171.7, 187.3). Which of the following statements are true about a 95% prediction interval for mortality rate at $x = 35$? **Select ALL that apply.**
- ☒ The center of the PI is the center of (171.7, 187.3).
 - ☐ The center of the PI is different from the center of (171.7, 187.3).
 - ☐ The PI is narrower than (171.7, 187.3).
 - ☒ The PI is wider than (171.7, 187.3).

7. (6 points) The HR director at a company plans to ask a sample of employees about whether they are satisfied with their job. He will use the sample to make a 95% CI for p , the true proportion of employees that are satisfied.

Assume the director is using the Wald standard error $\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ and is using critical value 1.96. How big of a sample size n does he need to take so that the margin of error of the interval will be **at most** 0.02?

Write an expression and then solve for n . You do not have to report the specific value of n but n should be the only unknown.

The standard error depends on $\hat{p}$, and it is largest when $\hat{p} = 0.5$. So, the required n is given by:

$$0.02 = 1.96 \sqrt{\frac{0.5 \times 0.5}{n}}$$

$$\left(\frac{0.02}{1.96}\right)^2 = \frac{0.25}{n}$$

$$n = \frac{0.25}{\left(\frac{0.02}{1.96}\right)^2}$$

8. 30 study participants were divided into groups of 15 and randomly selected to watch either a funny video or a sad video. Afterwards, each participant was asked if they wanted to give away half of their compensation money to a charity unrelated to the video they watched.

Let p_{funny} be the true proportion of people who view the funny video who choose to give money away. Let p_{sad} be the true proportion of people who view the sad video who choose to give money away. A Z test of hypotheses

$$H_0 : p_{funny} = p_{sad} \quad \text{versus} \quad H_A : p_{funny} > p_{sad}$$

resulted in a p-value of 0.8.

- (a) (4 points) Report the p-value for the following pairs of hypotheses.

i. $H_0 : p_{funny} = p_{sad}$ versus $H_A : p_{funny} < p_{sad}$
The p-value is $1 - 0.8 = 0.2$.

ii. $H_0 : p_{funny} = p_{sad}$ versus $H_A : p_{funny} \neq p_{sad}$
The p-value is $2 \times 0.2 = 0.4$.

- (b) (4 points) Which of the following statements about the study are true? **Select ALL that apply.**

- ☐ More participants who watched the funny video decided to give money away.
- ☒ More participants who watched the sad video decided to give money away.
- ☒ We do not have evidence at the 5% level that the groups have a different probability of giving money away.
- ☐ We have evidence at the 5% level that the groups have a different probability of giving money away.

9. A grocery store is concerned that chocolate bars from a supplier weigh less than the advertised 55 grams.

(a) (3 points) What are the null and alternative hypotheses that can be used to test μ , the true average weight of the chocolate bars? **Select ONE.**

- | | | | | | |
|---|--------|---------------------|---|--------|------------------|
| ☐ $H_0 : \mu = 55$ | versus | $H_A : \mu \neq 55$ | ☐ $H_0 : \mu \neq 55$ | versus | $H_A : \mu = 55$ |
| ☐ $H_0 : \mu = 55$ | versus | $H_A : \mu > 55$ | ☐ $H_0 : \mu > 55$ | versus | $H_A : \mu = 55$ |
| ☒ $H_0 : \mu = 55$ | versus | $H_A : \mu < 55$ | ☐ $H_0 : \mu < 55$ | versus | $H_A : \mu = 55$ |

(b) (4 points) The store takes a random sample of 40 of these chocolate bars and measure an average weight of 53 grams with standard deviation 2.5 grams.

Write a numerical expression that calculates the test statistic for your hypotheses. Do not simplify, evaluate, or round.

$$t_{obs} = \frac{53 - 55}{2.5/\sqrt{40}}$$

(c) (3 points) The store takes a different sample of 40 chocolate bars from a different brand and weighs them. The weights of the original 40 observations are in the R vector `brand1` and the weights of the new 40 observations are in the R vector `brand2`. Which R code can be used to test whether there is a difference in average weight between the two brands? **Select ONE.**

- ☒ `t.test(brand1, brand2)`
- ☐ `t.test(brand2, mu = 55)`
- ☐ `t.test(brand1, brand2, alternative = "greater")`
- ☐ `t.test(brand1, brand2, alternative = "less")`
- ☐ `t.test(brand1, brand2, paired = T)`